A Riesz representer perspective on targeted learning

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Causal Inference
Modus Operandi

1. Choose an estimand ψ and gather data $O_1, \ldots, O_n \sim \mathsf{P}$

2. Construct an estimator of ψ

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Targeted Learning

Even if P totally unknown (nonparametric), can construct "good" plug-in estimator of ψ by...

1. Deriving an "efficient influence function" $\phi(\mathsf{P})(O_i; \psi)$

2. Setting up and optimizing loss $L(O, \varepsilon)$ satisfying, for some vector v,

$$v^\top \nabla_\varepsilon L(O; \varepsilon)\Big|_{\varepsilon=0} = \frac{1}{n}\sum_{i=1}^{n} \phi(\mathsf{P})(O_i; \psi)$$

Choose L to respect problem constraints (i.e. bounded outcomes)

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TMLE.jl

Search docs (Ctrl + /)

R/`txshift`

Efficient Estimation of the Causal

R/`tmle3mediate`

Targeted Learning for Causal Mediation Analysis

Journal of Statistical Software

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Itmle: An R Package Implementing Targeted Minimum Loss-Based Estimation for Longitudinal Data

Samuel D. Lendle, Joshua Schwab, Maya L. Petersen, Mark J. van der Laan

Authors: Nima Hejazi, James Duncan, David McCoy, and Mark van der Laan

Q: Can we derive an EIF for a *general* class of estimands?

Estimand Name	**Formula for ψ**
Counterfactual mean (Static)	$\mathrm{E}[\mathrm{E}(Y \mid A = a, L)]$
Counterfactual mean (Stochastic)	$\mathrm{E}[\mathrm{E}(Y \mid A = A^{\star}, L)]$
Average quantile effect	$\mathrm{E}[Q_{\tau}(Y \mid A = 1, L) - Q_{\tau}(Y \mid A = 0, L)]$
Average treatment effect on treated	$\mathrm{E}(\mathrm{E}(Y \mid A = 1, L) \mid A = 1)$
Mediational effect	$\mathrm{E}[\mathrm{E}(\mathrm{E}(Y \mid A = 1, M, L) \mid A = 0, L)]$
Longitudinal treatment-specific mean	$\mathrm{E}[\mathrm{E}(\ldots \mathrm{E}(Y \mid \bar{A}_t = \bar{a}_T, \bar{L}_T) \ldots \mid A_1 = a_1, L_1, L_0)]$
Longitudinal modified treatment effect	$\mathrm{E}[\mathrm{E}(\ldots \mathrm{E}(Y \mid \bar{A}_t = \bar{A}_t^d, \bar{L}_T) \ldots \mid A_1 = A_1^d, L_1, L_0)]$

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Estimand Name	**Formula for ψ**
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Counterfactual mean (Stochastic)	
Average quantile effect	
Average treatment effect on treated	$\mathrm{E}[h(O;\eta)]$
Mediational effect	
Longitudinal treatment-specific mean	
Longitudinal modified treatment effect	

data

$$\mathbb{E}[h(O;\eta)]$$

function "evaluator"

nuisance function

Riesz Representation

Riesz Representation Theorem (statistics version).

Suppose $\eta \in \mathcal{L}_2(\mathsf{P})$, and $\psi := \Psi(\eta) = \mathsf{E}(h(O;\eta))$ is a bounded linear functional. Then, there exists $\alpha \in \mathcal{L}_2(\mathsf{P})$ such that

$$\Psi(\eta) = \mathsf{E}[\alpha(O)\eta(O)]$$

Think of α like a "balancing weight"!

Theorem (Riesz EIF). The efficient influence function of $\Psi(\eta)$ is

$$\phi(\mathsf{P})(O) = \underbrace{h(O;\eta_{\mathsf{P}}) - \Psi(\eta_{\mathsf{P}})}_{\text{expected value EIF}} + \underbrace{\int \alpha(O)\phi_\eta(\mathsf{P})(O)d\mathsf{P}}_{\text{"reweighted nuisance bias"}}$$

where $\phi_\eta(\mathsf{P})(O)$ denotes the efficient influence function of the nuisance parameter η.

Generalizes previous work, like Hirshberg and Wager (2021), Chernozhukov et al. (2022), or Williams et al. (2025)

Corollary (Riesz EIF with conditional mean nuisance). Suppose the observed data is $O = (L, A, Y)$, and $\eta(A, L) = E(Y \mid A, L)$.

Then, the efficient influence function of $\Psi(\eta)$ is

$$\phi(\mathsf{P})(O) = h(A, L; \eta_{\mathsf{P}}) - \Psi(\eta_{\mathsf{P}}) + \alpha(A, L)(Y - \eta(A, L))$$

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Proof. Just plug in to previous theorem the known efficient influence function of conditional mean

$$\phi_\eta(O) = \frac{\delta_{A,L}}{d\mathsf{P}(A, L)}(Y - E(Y \mid A, L))$$

Side note: Also works with other nuisances. Consider instead treatment-specific quantile effect $\Psi(\eta) = E[Q^\tau(a, L)]$*, where* $\eta(O) = Q^\tau(A, L)$*. By the same logic, the EIF of* $\Psi(\eta)$ *is*

$$\phi(P)(O) = h(O; Q^\tau) - \psi + \alpha(A, L) \underbrace{\left(\frac{\tau - \backslash 1(Y > Q^\tau(A, L))}{d\mathsf{P}(Q^\tau(A, L) \mid A, L)} \right)}_{\text{reweighted "pinball loss" derivative}}$$

Many more ways to use this

Consider a time-ordered data structure

$$O = (L_1, A_1, \dots, L_T, A_T, Y)$$

Denote the histories at time t as $\bar{A}_t$ and $\bar{L}_t$. Encompasses:

- Longitudinal data
- Mediation
- Multi-phase sampling

Theorem (Sequential Riesz EIF)

Consider the estimand $\Psi(\eta_1) = \mathsf{E}_{\mathsf{P}}[h_1(\bar{A}_1, \bar{L}_1; \eta_1)]$, where η_t is a bounded linear functional defined sequentially such that, for $t = 1, \dots, T$, we have

$$\eta_t(\bar{A}_t, \bar{L}_t) = \mathsf{E}[h_{t+1}(\bar{A}_{t+1}, \bar{L}_{t+1}; \eta_{t+1}) \mid \bar{A}_t, \bar{L}_t]$$

with $h_T(\bar{A}_{T+1}, \bar{L}_{T+1}; \eta_{T+1}) := Y$. Let α_t denote the Riesz representer for η_{t+1} in the functional $\mathsf{E}[h_{t+1}(\bar{A}_{t+1}, \bar{L}_{t+1}; \eta_{t+1}) \mid \bar{A}_t, \bar{L}_t]$. Then, the EIF the estimand $\Psi(\mathsf{P}_0)$ is

$$\underbrace{h_1(\bar{A}_1, \bar{L}_1; \eta_1) - \psi_0}_{\text{Expected value EIF}} + \sum_{t=1}^{T} \underbrace{\prod_{k=1}^{t} \alpha_t(\bar{A}_t, \bar{L}_t)}_{\text{reweighting}} \underbrace{[h_{t+1}(\bar{A}_{t+1}, \bar{L}_{t+1}; \eta_{t+1}) - \eta_t(\bar{A}_t, \bar{L}_t)]}_{\text{Residuals of sequential regressions}}$$

Sequential TMLE

1. Fit sequential regressions $\eta_1, \dots, \eta_T$ and Riesz representers $\alpha_1, \dots, \alpha_T$.
2. For $t = 1, \dots T$, compute the weights $\omega_t(\bar{A}_{it}, \bar{L}_{it}) = \prod_{k=1}^{t} \alpha_k(\bar{A}_{ik}, \bar{L}_{ik})$.
3. Set $h_T(\bar{A}_{iT}, \bar{L}_{iT}; \eta_{iT,\hat{\varepsilon}_T}) = Y_i$
4. For $t = T-1, \dots, 1$, fit 1-D parametric model $\eta_{t,\hat{\varepsilon}_t}$ that regresses
$$\text{link}[\underbrace{h_{t+1}(\bar{A}_{i(t+1)}, \bar{L}_{i(t+1)}, \eta_{(t+1),\hat{\varepsilon}_{(t+1)}})}_{\text{outcome } \eta_t \text{ was fitted on}}] = \text{link}[\underbrace{\eta_t(\bar{A}_{it}, \bar{L}_{it})}_{\substack{\text{offset: original} \\ \text{regression}}}] + \underbrace{\varepsilon_t \omega_t}_{\substack{\text{clever} \\ \text{covariate}}}$$
and set $\eta_t = \eta_{t,\hat{\varepsilon}_t}$
5. The final TMLE update is the updated plug-in estimator

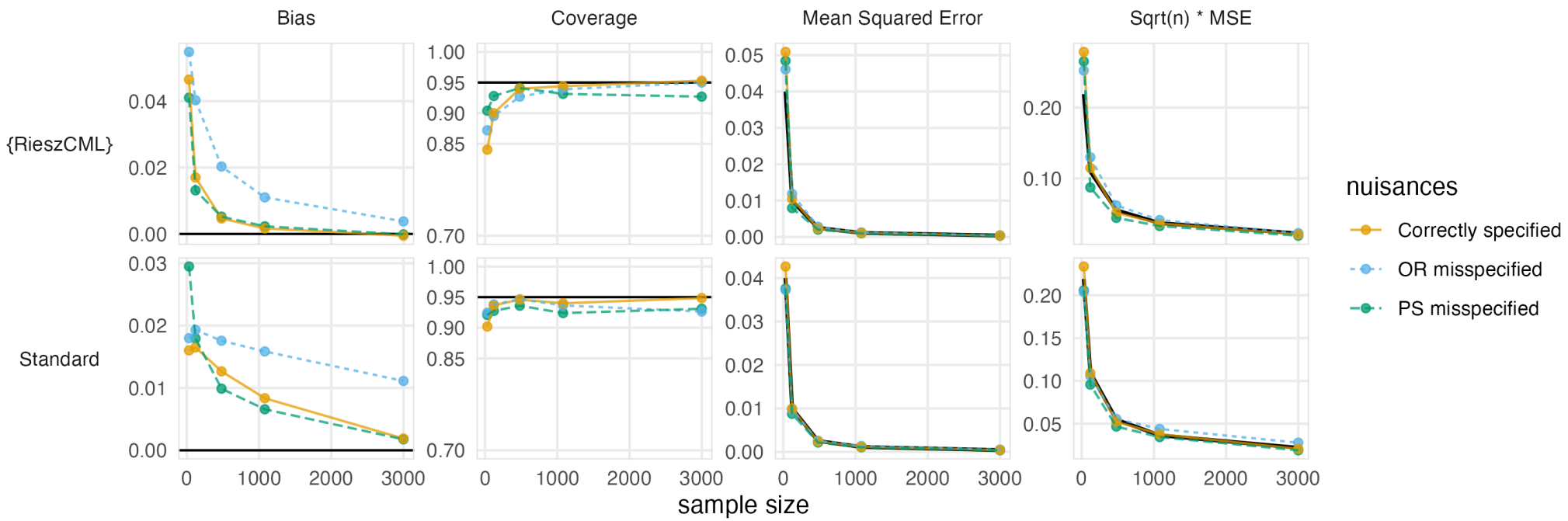
$$\psi_n = \frac{1}{n} \sum_{i=1}^{n} \text{link}[h_1(\bar{A}_{i1}, \bar{L}_{i1}; \eta_{1,\hat{\varepsilon}_1})]$$

→ *Consistent, asymptotically normal, and semi-parametric efficient*

The `RieszCML` package: Simulations

Convergence diagnostics for {RieszCML} TMLE and standard software

Black lines indicate ideal bias, coverage, and the efficiency bound.



All experiments were run for 2,000 replications.

Conclusions

- Unifies semi-parametric efficient estimation across…
  - → *Data*: Longitudinal, mediation, multi-phase sampling, etc.
  - → *Interventions*: Binary, stochastic, derivatives, quantile effects, etc.
- Support theory and software re-use (*Lego bricks*)
- Agnostic about how α is learned; see, e.g. *Riesz regression* (Chernozhukov *et al.*, 2022)

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Thank you! Questions?

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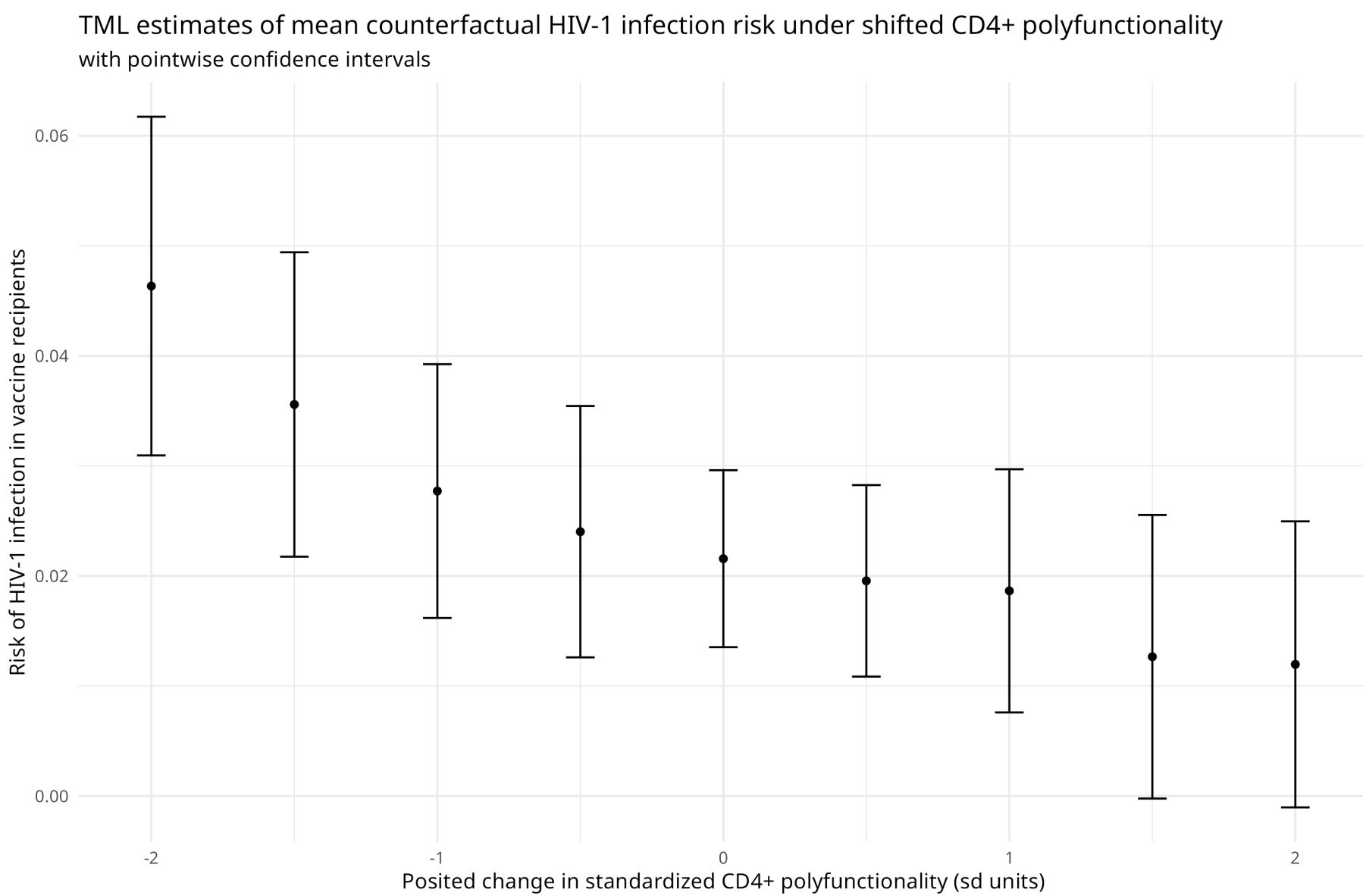
Appendix A: Data Analysis



Appendix B: Functional analysis

Theorem (Riesz Representation, general).

Suppose $\eta \in \mathcal{H}$, a Hilbert space, and that $\psi(\eta) : \mathcal{H} \mapsto \mathbb{R}$ is a bounded linear functional.

Then, there exists $\alpha \in \mathcal{H}$ such that

$$\psi(\eta) = \langle \alpha, \eta \rangle$$